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Math 208 Final Exam Term 252

Original exam date: 12 May 2026. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ has an eigenvalue $\lambda = 1$ of defect 2. If we choose $V_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ such that $(A - I)^3 V_3 = 0$ and $(A - I)^2 V_3 \neq 0$, then the general solution of the system $X' = AX$ is

(a) $X = \left(c_1 \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2t \\ 1+2t \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ t^2+t \\ t^2 \end{bmatrix} \right) e^t$

(b) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 2t+2 \\ 2t+2 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ t^2+2t \\ t^2+2t \end{bmatrix} \right) e^t$

(c) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ 2t \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ t^2+2t \\ t^2+2t \end{bmatrix} \right) e^t$

(d) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2t+1 \\ 2t+1 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ t^2+2t \\ t^2+2t \end{bmatrix} \right) e^t$

(e) $X = \left(c_1 \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 2t+2 \\ 2t \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ t^2+2t \\ t^2 \end{bmatrix} \right) e^t$

2. If the recurrence relation of the Frobenius series solution of the differential equation

$$2x^2y'' - 3xy' + (3+x)y = 0$$

that corresponds to the indicial root $r = \dfrac{3}{2}$ is given by $c_0 = 1$, $c_n = -\dfrac{1}{n(2n+1)} c_{n-1}$, $n \geq 1$, then the first three terms in the solution are given by

(a) $x^{3/2} - \dfrac{1}{3}x^{5/2} - \dfrac{1}{15}x^{7/2}$

(b) $x^{3/2} - \dfrac{1}{5}x^{5/2} + \dfrac{1}{30}x^{7/2}$

(c) $x^{3/2} - \dfrac{1}{3}x^{5/2} + \dfrac{1}{30}x^{7/2}$

(d) $x^{3/2} - \dfrac{1}{3}x^{5/2} - \dfrac{7}{30}x^{7/2}$

(e) $x^{3/2} - \dfrac{1}{4}x^{5/2} + \dfrac{1}{30}x^{7/2}$

3. By using the method of undetermined coefficients, a particular solution of the differential equation $y'' - 3y' = 8e^{3x} + 4\sin x$ is given by

$$y_p = Axe^{3x} + B\cos x + C\sin x,$$

where $5B + 3A + 5C =$

(a) 8

(b) 12

(c) 10

(d) 0

(e) 4

4. Consider the non-homogeneous system $X' = AX + \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. If the general solution of the associated homogeneous system is

$$X_c = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -4 \\ 6 \end{bmatrix} e^{2t},$$

then a particular solution $X_p =$

(a) $\begin{bmatrix} -9 \\ 6 \end{bmatrix}$

(b) $\begin{bmatrix} -9 \\ 4 \end{bmatrix}$

(c) $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$

(d) $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$

(e) $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$

5. The solution of the initial-value problem $xy^2\dfrac{dy}{dx} = y^3 - x^3$, $y(1) = 3$ is given by

(a) $y^3 + 2x^3 \ln|x| = 27x^3$

(b) $3y^2 + 3x^3 \ln|x| = 27x^3$

(c) $y^3 + 3x^3 \ln|x| = 27x^3$

(d) $y^3 - 4x^3 \ln|x| = 27x^3$

(e) $9y + 3x^3 \ln|x| = 27x^3$

6. A particular solution of the differential equation $y'' + y = \sec^2 x$ is given by $y_p(x) =$

(a) $\cos x + x \ln|\sec x + \tan x|$

(b) $-1 + \sin x \ln|\sec x + \tan x|$

(c) $x + \sin x \ln|\sec x + \tan x|$

(d) $x \cos x + \sin x \ln|\sec x + \tan x|$

(e) $x^2 + \cos x \ln|\sec x + \tan x|$

7. The general solution of the differential equation

$$\frac{dy}{dx} + y \tan x = \cos^2 x$$

is given by

(a) $y = (2 \sin x + c) \cos^2 x$

(b) $y = (\sin x + c) \cos x$

(c) $y = (\sin^2 x + c) \cos x$

(d) $y = (\cos^2 x + c) \sin x$

(e) $y = (\cos x + c) \sin x$

8. If $K = \begin{bmatrix} 1 \\ \alpha \\ 0 \end{bmatrix}$ is an eigenvector with eigenvalue $\lambda = 2$ of the matrix $A = \begin{bmatrix} 0 & -2 & -1 \\ -3 & -1 & -3 \\ 2 & 2 & 3 \end{bmatrix}$, then $\alpha =$

(a) 0

(b) -2

(c) 2

(d) -1

(e) 1

9. For $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$, if P is a diagonalizing matrix such that $P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix}$, then

(a) $P = \begin{bmatrix} 1 & 3 \\ -1 & 3 \end{bmatrix}$

(b) $P = \begin{bmatrix} -1 & 2 \\ 1 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$

10. If $A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}$, then $e^{At} = \begin{bmatrix} e^t & f(t) & h(t) \\ 0 & e^t & g(t) \\ 0 & 0 & e^t \end{bmatrix}$, where $f(1) + g(1) + h(1) =$

(a) $20\,e$

(b) $16\,e$

(c) $14\,e$

(d) $17\,e$

(e) $22\,e$

11. The number of regular singular points of the differential equation

$$x^3(x^2 - 25)(x - 2)^2y'' + 3x(x - 2)y' + 7(x + 5)y = 0$$

is

(a) 1

(b) 3

(c) 4

(d) 0

(e) 2

12. A fundamental matrix of the system $X' = \begin{bmatrix} 4 & 0 \\ 1 & -1 \end{bmatrix} X$ is

(a) $\begin{bmatrix} 3e^{4t} & 0 \\ 0 & e^{-t} \end{bmatrix}$

(b) $\begin{bmatrix} e^{4t} & 0 \\ 2e^{4t} & e^{-t} \end{bmatrix}$

(c) $\begin{bmatrix} 5e^{4t} & 0 \\ e^{4t} & e^{-t} \end{bmatrix}$

(d) $\begin{bmatrix} 5e^{4t} & 0 \\ 2e^{4t} & e^{-t} \end{bmatrix}$

(e) $\begin{bmatrix} 5e^{4t} & e^{-t} \\ e^{4t} & e^{-t} \end{bmatrix}$

13. If the rank of the matrix

$$\begin{bmatrix} 1 & 1 & 3 & 3 & 0 \\ -1 & 0 & -2 & -1 & 1 \\ 2 & 3 & 7 & 8 & K \\ -2 & 4 & 0 & 6 & 6 \end{bmatrix}$$

is equal to 2, then $K =$

(a) 2

(b) 3

(c) 1

(d) 0

(e) -2

14. The general solution of the system

$$X' = \begin{bmatrix} 10 & -5 \\ 8 & -12 \end{bmatrix} X$$

is given by

(a) $X = c_1 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{6t}$

(b) $X = c_1 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-10t}$

(c) $X = c_1 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{10t}$

(d) $X = c_1 \begin{bmatrix} 5 \\ 1 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{-10t}$

(e) $X = c_1 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{-10t}$

15. Consider the system $X' = AX$, $X(0) = \begin{bmatrix} -2 \\ 8 \end{bmatrix}$, where A is a 2×2 matrix with real entries. If A has an eigenvalue $\lambda = 5 + 2i$ with corresponding eigenvector $K = \begin{bmatrix} -1 \\ 1 - 2i \end{bmatrix}$, then $X\left(\frac{\pi}{2}\right) =$

(a) $\begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{\frac{5\pi}{2}}$

(b) $\begin{bmatrix} 2 \\ -8 \end{bmatrix} e^{\frac{5\pi}{2}}$

(c) $\begin{bmatrix} -1 \\ 8 \end{bmatrix} e^{\frac{5\pi}{2}}$

(d) $\begin{bmatrix} -1 \\ 4 \end{bmatrix} e^{\frac{5\pi}{2}}$

(e) $\begin{bmatrix} 2 \\ -6 \end{bmatrix} e^{\frac{5\pi}{2}}$

16. The solution of the exact differential equation

$$(1 + 2x - y^3)\,dx + (2y - 3xy^2)\,dy = 0$$

is given by

(a) $x + x^2 + 3xy^3 + y^2 = c$

(b) $x + x^2 - xy^3 + y^2 = c$

(c) $x^2 + x^3 - xy^3 + y^2 = c$

(d) $x^2 + x^3 - xy^3 + 4y^2 = c$

(e) $x + x^2 + 2xy^3 + y^2 = c$

17. The largest indicial root at $x = 0$ for the differential equation

$$8xy'' + y' + 2y = 0$$

is

(a) $\frac{1}{2}$

(b) 0

(c) $\frac{7}{8}$

(d) $\frac{3}{4}$

(e) $\frac{5}{8}$

18. The general solution of the differential equation $(D-1)^2(D^2-2D+5)y=0$ is given by

(a) $y(x)=c_1e^{-x}+c_2xe^{-x}+c_3e^{-2x}\cos x+c_5e^{-2x}\sin x$

(b) $y(x)=c_1e^{x}+c_2xe^{x}+c_3e^{x}\cos(2x)+c_4e^{x}\sin(2x)$

(c) $y(x)=c_1e^{x}+c_2xe^{x}+c_3\cos(2x)+c_4\sin(2x)$

(d) $y(x)=c_1e^{-x}+c_2xe^{-x}+c_3e^{x}\cos(2x)+c_4e^{x}\sin(2x)$

(e) $y(x)=c_1e^{x}+c_2xe^{x}+c_3e^{-x}\cos(2x)+c_4e^{-x}\sin(2x)$

19. If $y=\sum_{n=0}^{\infty}c_nx^n$ is a power series solution of the differential equation $(x^2+1)y''+xy'-y=0$ about the ordinary point $x=0$, then the constants c_n satisfy the recurrence relation:

(a) $c_{n+1}=\dfrac{1-n}{n+2}c_n,\ n\geq 2$

(b) $c_{n+2}=\dfrac{2-n}{n+2}c_n,\ n\geq 3$

(c) $c_{n+2}=\dfrac{3-n}{n+1}c_n,\ n\geq 4$

(d) $c_{n+2}=\dfrac{1-n}{n+2}c_n,\ n\geq 2$

(e) $c_{n+1}=\dfrac{2-n}{n+2}c_n,\ n\geq 2$

20. The general solution of $X'=\begin{bmatrix}9&1&1\\1&9&1\\1&1&9\end{bmatrix}X$ can be written as

$$X=c_1\begin{bmatrix}1\\1\\1\end{bmatrix}e^{11t}+c_2\begin{bmatrix}1\\\beta\\0\end{bmatrix}e^{8t}+c_3\begin{bmatrix}1\\0\\\alpha\end{bmatrix}e^{8t},$$

then $\alpha-\beta=$

(a) 0

(b) 2

(c) -2

(d) 3

(e) -3

21. If the three vectors $V_1 = (4, 3, 2)$, $V_2 = (2, 5, -1)$ and $V_3 = (k, -4, 2)$ of $\mathbb{R}^3$ are linearly dependent, then $k =$

(a) $\dfrac{3}{11}$

(b) $-\dfrac{5}{13}$

(c) $-\dfrac{4}{11}$

(d) $-\dfrac{4}{13}$

(e) $\dfrac{5}{13}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Chain $V_1 = (0,2,2)$, $V_2 = (0,2,0)$, $V_3 = (1,0,0)$ gives X_1, X_2, X_3 times e^t.
2	**(c)**	$c_1 = -\frac{1}{3}$, $c_2 = \frac{1}{30}$, so $x^{3/2} - \frac{1}{3}x^{5/2} + \frac{1}{30}x^{7/2}$.
3	**(b)**	$3A = 8$, $B = \frac{6}{5}$, $C = -\frac{2}{5}$, so $5B + 3A + 5C = 12$.
4	**(a)**	Variation of parameters: $\Phi^{-1}\mathbf{f} = (15e^{-t}, 3e^{-2t})$, $X_p = (-9, 6)$.
5	**(c)**	$v = y^3$: $v' - \frac{3}{x}v = -3x^2$, $\rho = x^{-3}$, giving $y^3 + 3x^3 \ln\lvert x\rvert = 27x^3$.
6	**(b)**	$u_1 = -\sec x$, $u_2 = \ln\lvert\sec x + \tan x\rvert$: $y_p = -1 + \sin x \ln\lvert\sec x + \tan x\rvert$.
7	**(b)**	$\rho = \sec x$: $(y \sec x)' = \cos x$, so $y = (\sin x + c)\cos x$.
8	**(d)**	$AK = 2K$: first entry $-2\alpha = 2$, so $\alpha = -1$ (other entries agree).
9	**(e)**	$\lambda = -1 : (1,-1)$, $\lambda = 4 : (3,2)$, so $P = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$.
10	**(e)**	$e^{At} = e^t(I + Nt + N^2t^2/2)$: $f = 3te^t$, $g = 6te^t$, $h = (4t + 9t^2)e^t$.
11	**(b)**	$x = 0$ is irregular; $x = 5, -5, 2$ are regular, so 3.
12	**(c)**	Eigenpairs $(4, (5,1))$ and $(-1, (0,1))$ give $\Phi = \begin{bmatrix} 5e^{4t} & 0 \\ e^{4t} & e^{-t} \end{bmatrix}$.
13	**(c)**	Reduced rows $(0,1,1,2,1)$ and $(0,1,1,2,K)$ must agree, so $K = 1$.
14	**(e)**	$\lambda = 8, -10$ with eigenvectors $(5,2)$ and $(1,4)$.
15	**(b)**	$c_1 = 2$, $c_2 = -3$; at $t = \pi/2$: $X = (2,-8)e^{5\pi/2}$.
16	**(b)**	Exact: $F = x + x^2 - xy^3 + g(y)$, $g' = 2y$, so $x + x^2 - xy^3 + y^2 = c$.
17	**(c)**	$r(r-1) + \frac{1}{8}r = 0 \Rightarrow r = 0, \frac{7}{8}$; largest is $\frac{7}{8}$.
18	**(b)**	Roots $1, 1, 1 \pm 2i$: $y = (c_1 + c_2x)e^x + e^x(c_3\cos 2x + c_4 \sin 2x)$.
19	**(d)**	$(n+2)(n+1)c_{n+2} = (1-n^2)c_n$, so $c_{n+2} = \frac{1-n}{n+2}c_n$.
20	**(a)**	$A - 8I$ is all ones, so $\beta = -1$, $\alpha = -1$ and $\alpha - \beta = 0$.
21	**(d)**	$\det = -4 - 13k = 0$, so $k = -\frac{4}{13}$.